A Theorem on the Imaginary Number $\sqrt{-1}$

Chinnaraji Annamalai
School of Management, Indian Institute of Technology, Kharagpur, India
Email: anna@iitkgp.ac.in
https://orcid.org/0000-0002-0992-2584

$\sqrt{-1}$ is an imaginary number in the complex number system. The paper presents a theorem using the imaginary number $\sqrt{-1}$**.**

Theorem: $\sqrt{-1} = \frac{1}{\sqrt{-1}} \Rightarrow \left(\sqrt{-1}\right)^n = \frac{1}{\left(\sqrt{-1}\right)^n}$, where the integer $n \geq 1$.

Proof.

Method I

$\sqrt{-1}$ is an imaginary number in the complex number system [1].

$$\sqrt{-1} = \frac{1}{\sqrt{-1}} \Rightarrow \sqrt{-1} = \frac{\sqrt{(-1)\times(-1)}}{\sqrt{-1}} \Rightarrow \sqrt{-1} = \frac{\sqrt{-1}\times\sqrt{-1}}{\sqrt{-1}} = \sqrt{-1}.$$

From this result, we conclude that

$$\frac{1}{\left(\sqrt{-1}\right)^n} = \frac{\left(\sqrt{-1}\times\sqrt{-1}\right)^n}{\left(\sqrt{-1}\right)^n} = \frac{\left(\sqrt{-1}\right)^n \times \left(\sqrt{-1}\right)^n}{\left(\sqrt{-1}\right)^n} = \left(\sqrt{-1}\right)^n.$$

$$\therefore \sqrt{-1} = \frac{1}{\sqrt{-1}} \Rightarrow \left(\sqrt{-1}\right)^n = \frac{1}{\left(\sqrt{-1}\right)^n}.$$

Method II

By multiplying the imaginary number $\sqrt{-1}$ on both sides of $\sqrt{-1} = \frac{1}{\sqrt{-1}}$, we get

$$\sqrt{-1}\sqrt{-1} = \frac{\sqrt{-1}}{\sqrt{-1}} \Rightarrow \sqrt{(-1)\times(-1)} = 1 \Rightarrow \sqrt{1} = 1 \Rightarrow 1 = 1.$$

$$\left(\sqrt{-1}\right)^2 = \frac{\sqrt{-1}}{\sqrt{-1}} \Rightarrow \sqrt{-1}\sqrt{-1} = 1 \Rightarrow 1 = 1, \text{ i. e. } \left(\sqrt{-1}\right)^2 = 1.$$

Hence, $\sqrt{-1} = \frac{1}{\sqrt{-1}}$ is proved.

Let $n = 3$ in $\left(\sqrt{-1}\right)^n = \frac{1}{\left(\sqrt{-1}\right)^n}$. Then, $\left(\sqrt{-1}\right)^3 = \frac{1}{\left(\sqrt{-1}\right)^3} \Rightarrow \left(\sqrt{-1}\right)^6 = \left[\left(\sqrt{-1}\right)^2\right]^3 = 1,$

where $\left(\sqrt{-1}\right)^2 = 1$.

Hence, $\left(\sqrt{-1}\right)^3 = \frac{1}{\left(\sqrt{-1}\right)^3}$ is proved.

Similarly, we can prove that $\left(\sqrt{-1}\right)^n = \frac{1}{\left(\sqrt{-1}\right)^n}$.

$\left(\sqrt{-1}\right)^n = \frac{1}{\left(\sqrt{-1}\right)^n} \Rightarrow \left(\sqrt{-1}\right)^{2n} = \left[\left(\sqrt{-1}\right)^2\right]^n = 1 \Rightarrow (1)^n = 1 \Rightarrow 1 = 1.$

$\therefore \sqrt{-1} = \frac{1}{\sqrt{-1}} \Rightarrow \left(\sqrt{-1}\right)^n = \frac{1}{\left(\sqrt{-1}\right)^n}$.

Hence, theorem is proved.

References

[1] Wikipedia. https://en.wikipedia.org/wiki/Complex_number.